# DRIVING DEMAND

An AI-Backed Driver Nudge System for Demand-Aware Positioning Decision

***Ride-hail apps tell drivers WHERE to go. We tell them WHY — and let them decide.***

**9.4M**
YELLOW TAXI TRIPS · JAN–MAR 2023

TEAM 7
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# THE PROBLEM

*Ride-hailing platforms suffer from a critical supply–demand mismatch.*

### DRIVER IDLE TIME

Drivers wait in the wrong zones, losing earnings while riders queue up across the city.

### ZERO VISIBILITY

No reasoning behind dispatch, drivers only see the order to move, never why a zone matters.

### LOST REVENUE

Mismatches cause longer waits, cancellations, and fewer completed rides per driver hour.

# OUR SOLUTION

A demand-aware driver decision-support system, built on big data infrastructure and grounded LLM reasoning.

SCALE

**9.4M**

**NYC YELLOW TAXI TRIPS**

Jan–Mar 2023 · NYC TLC open data joined with the Taxi Zone lookup (263 zones across 5 boroughs).

PySpark · Delta Lake · Unity Catalog

### LARGE-SCALE PROCESSING

11-part PySpark pipeline writing five Delta Lake tables.

### ZONE DEMAND SCORING

Zone trips ÷ citywide average · low-confidence cells filtered out.

### LLM NUDGE GENERATION

8 windows · few-shot prompting · strict grounding rules.

### VALIDATED RELIABILITY

Temporal holdout vs a 20% random baseline.

# DEMAND INTELLIGENCE PIPELINE

*Seven analysis phases, five Delta tables, from raw trip data to validated nudges.*

**01 INGEST**
Load 3 monthly Parquet files + zone lookup

→

**02 CLEAN**
Dates · zones 1–263 · fare & distance outliers

→

**03 AGGREGATE**
Zone × day-type × daypart

→

**04 SCORE**
zone ÷ citywide avg · flag low-confidence

→

**05 RANK**
Top-5 zones per window × 8 windows

MORNING_RUSH · 6–9:59 | MIDDAY · 10–15:59 | EVENING_RUSH · 16–19:59 | NIGHT · 20–5:59 | × WEEKDAY / WEEKEND

**06 NUDGE GENERATION**
OpenAI Chat API · few-shot prompt · one grounded message per window.

**07 VALIDATION**
Temporal holdout hit-rate against real March 2023 demand.

# EXPLAINABLE NUDGES

Structured demand signals → LLM → clear, trustworthy driver guidance.

1. **DEMAND RANKING**
   Confidence filter, then Top-5 zones for each of the 8 windows.

2. **FEW-SHOT PROMPTING**
   Gold examples fix tone, length, and grounding before every API call.

3. **NUDGE GENERATED**
   One short, grounded message per window, produced by the Chat API.

4. **QUALITY GUARDRAILS**
   ≤ 60 words · ≤ 2 zones · ~Nx multipliers · no “surge / traffic / earnings.”

**★ EXAMPLE OUTPUT**

WEEKEND · EVENING RUSH

*“During the weekend evening rush, JFK Airport is about **14.9x** busier than average with roughly 1,200 pickups per evening. East Village follows at about **12.7x** with around 1,000 pickups per evening. **Focus on JFK Airport for potential opportunities.** “*

< 60 words | zones cited | grounded

# EVALUATION & RESULTS

*Temporal holdout validation, predictions tested against real March demand.*

STEP 01
**Train on January–February trip patterns**

→

STEP 02
**Rank top zones per time window**

→

STEP 03
**Label March zones by demand percentile**

→

STEP 04
**Compare vs 20% random baseline**

HIT RATE · ALL 8 WINDOWS

**100%**

**5×** better than random chance (20%)

**BEATS 20% BASELINE**
Consistent outperformance across the holdout month.

**STABLE ACROSS 8 / 8 WINDOWS**
Weekday / weekend × four dayparts, no collapse.

**8 / 8 NUDGES PASSED**
Every generated message cleared all guardrail checks.

# KEY IMPACT

*What this system delivers for drivers and platforms.*

### REDUCE IDLE TIME

Guide drivers to high-demand zones before peaks form.

### MORE COMPLETED RIDES

Better positioning means more pickups per shift.

### DRIVER EARNINGS UP

Data-driven repositioning raises per-hour income.

### EXPLAINABLE AI

Drivers know **why** — building trust and higher adoption.

# DRIVING DEMAND

- 9.4M NYC Yellow Taxi trips processed end-to-end
- Normalized zone scoring with confidence filtering
- LLM nudges grounded in real data, not speculation
- **100%** hit rate on March holdout, 5× better than random

**100%**

HIT RATE · MARCH HOLDOUT

## Q&A

*Thank you!*